II श्री गणेश प्रसन्न II

CAMPBELL-WALSH-WEIN · BOOK 8

Nocturia & Male Incontinence

Concise rapid revision · facts answered in Papers 1–4 point to that answer

highlighted number to memorise · KEY POINT and ALGORITHM boxes carry the reasoning · P2 = covered in full in that Paper's answer book; the list under it gives what that answer misses

Chapter	Topic
Ch 112	Nocturia
Ch 123	Surgical Procedures for Sphincteric Incontinence in the Male: Artificial Urinary Sphincter and Perineal Slings
★	Putting it together

112 Nocturia

1 · Definition and terminology

- **Nocturia** (International Continence Society, ICS, 2019) is the number of voids during the **main sleep period**, from intending to sleep until intending to rise. The period follows the **individual's own schedule**, so shift workers are included.
- The definition does **not exclude 1 void per night** and does **not incorporate bother**.
- ICS 2019 dropped "wake **to void**"; non-urological causes are **leverage for interdisciplinary care, not grounds to exclude**.
- Guidance is from the European Association of Urology (EAU) and ICS; the AUA overactive bladder (OAB) guideline calls **isolated nocturia a separate entity**.

Key diary terms

Term	Definition / formula	Meaning
Actual nightly voids (ANV)	Voids in the main sleep period	Severity count
First uninterrupted sleep period (FUSP)	Hours from sleep onset to the first night void	Key sleep measure
Nocturnal urine volume (NUV)	Night voids **plus the first morning void**	Volume made during sleep
Nocturnal polyuria index (NPi)	NUV / 24-h volume	Proportion made at night
Nocturnal urine production (NUP)	NUV / hours of sleep	Hourly rate
Maximum voided volume (MVV); nocturnal MVV (NMVV)	Largest void in 24 h; largest in the sleep period	Proxies for functional capacity
Nocturia index (Ni)	NUV / MVV	>1: night output exceeds capacity
Modified Ni (MNi)	NUV / NMVV	Uses night data only
Predicted nightly voids (PNV)	Ni − 1	Voids expected if the bladder filled fully each time
Nocturnal bladder capacity index (NBCi)	ANV − PNV	>0: patient voids below capacity

Trap: the first morning void is **split**. Its volume goes into NUV, but it counts as daytime frequency, not nocturia.

2 · Why nocturia matters

- **It is common**: up to **3 in 5** people >70 years void ≥2 times nightly; age is the most robust correlate.
- **Bother** is reported by most people with ≥**2** voids, but **any nocturia that brings a patient to a urologist is clinically significant**.
- **Sleep**: nocturia is the **commonest cause of adult sleep disruption**. Most first voids occur within **2–3 h** of sleep onset, breaking early slow-wave (N3) sleep and causing daytime fatigue.
- **Falls and hip fracture** are linked to nocturia more than to other lower urinary tract symptoms (LUTS).

- **Systemic disease**: nocturia may herald cardiometabolic (strongest), renal, endocrine or neurological disease; ≥2 voids is linked to higher mortality.

> **KEY POINT**
> The first void breaks restorative slow-wave sleep; the symptom may be the visible edge of systemic disease.

3 · Pathophysiology — production, capacity and sleep

Remember **"Sleep CALM"**: **S**leep disorders, **C**omorbidities, **A**ctions (behaviour, lifestyle), **L**ower urinary tract dysfunction (LUTD) and **M**edications.

- Every factor upsets **night production**, **functional capacity** or **sleep**; the sleep link is **bidirectional** ("convenience" voids).

Nocturnal polyuria (NP) and global polyuria

Criterion	Threshold	Comment
NPi (ICS)	> **20%** in young adults to > **33%** in older adults	Most used, but discriminates poorly
NUP (ICS alternative)	> **90 mL/h**	More specific for production-driven nocturia
Global polyuria	> **40 mL/kg/24 h** (>2.8 L at 70 kg)	Uncommon but serious; **always investigate**

- **Krimpen study**: NPi >33% was found in 92% of older men with ≥2 voids, but also in 70% of men **without** nocturia.
- Excess night production contributes to **most adult nocturia**.

Functional capacity and the mismatch

- Normal capacity is usually >300–400 mL; **150–300 mL** suggests diminished capacity, which may be **24-hour** (all voids small) or **night-specific** (partly convenience voids).
- **Ni is the strongest diary determinant of severity and of treatment response**.

> **ALGORITHM**
> 1. Worked example:
> 2. NUV is 800 mL and MVV is 200 mL, so Ni = 4.
> 3. PNV = Ni − 1 = 3.
> 4. The diary shows 6 night voids, so NBCi = 6 − 3 = **3**: the patient voids well below capacity.

Investigate diminished night capacity when NBCi is > **1.3**; a value > **2** is strongly associated with severe nocturia.

4 · The bladder diary

EAU (strong): use a diary for female LUTS, and for male LUTS with storage symptoms or nocturia.

ALGORITHM

1. Give the patient a measuring container.
2. Record time and volume of every void and every drink.
3. Mark bedtime (intending to sleep) and rising time.
4. Grade urgency and note leakage for each void.

- International Consultation on Incontinence (ICI) urgency codes: **0** no need (social void), **1** normal desire, **2** urgency that passed, **3** urgency but dry, **4** urgency with leak.
- **Diary ANV is the gold standard for severity**; questionnaires overestimate it. **FUSP is the most important sleep measure**.
- Duration: EAU advises **3 days** (Male guideline: at least 3); because of "diary fatigue", fewer than half complete it.
- A **nocturnal-only diary** lets MNi substitute for Ni but **cannot diagnose global polyuria**; EAU does not recommend it for women.

KEY POINT

Every cause works through production against capacity, gated by sleep; their ratio (Ni) predicts severity better than either alone.

5 · Evaluation — the Sleep CALM work-up

Do a basic work-up in all, then tailored tests and referral. The **TANGO** questionnaire screens all domains.

Factor	History / examination	Tests (basic, then ancillary)
Sleep	Habits, apnoea, restless legs, pain, mood	STOP-BANG and sleep diary; then polysomnography, sleep specialist
Comorbidities	Blood pressure, volume overload, neurological signs	Metabolic panel; then HbA1c, ECG, echo, natriuretic peptide, urine albumin, calcium, thyroid, water deprivation
Actions	Fluid, alcohol, caffeine, sodium; body mass index (BMI)	Fluid intake diary
LUTD	LUTS, prior surgery; abdominal, genital, rectal examination	Urinalysis, post-void residual (PVR); then prostate-specific antigen (PSA), uroflowmetry, cystoscopy
Medications	Reconcile drugs and **dose timing**	—

- **Trap**: urology is one row of five; first-line is to **optimise every Sleep CALM abnormality**.

6 · Sleep disorders

Obstructive sleep apnoea (OSA) is the best-described link; frequent nocturia predicts severe OSA.

Arm	OSA mechanism	Effect of continuous positive airway pressure (CPAP)
Production (main)	Hypoxia causes pulmonary vasoconstriction, raising right atrial transmural pressure. This releases **atrial natriuretic peptide (ANP)**, causing natriuresis.	Lowers night production

Arm	OSA mechanism	Effect of continuous positive airway pressure (CPAP)
Capacity	Intermittent hypoxia causes oxidative stress and detrusor instability	Less detrusor overactivity
Sleep	More light sleep, less slow-wave sleep	Increases slow-wave sleep

- CPAP reduces nocturia by up to ≈**2 voids** in moderate–severe OSA, among the **largest effects of any intervention**. EAU Female (strong): refer suspected OSA to a sleep clinic.
- **STOP-BANG** scores Snoring, Tired, Observed apnoea, Pressure, BMI >**35**, Age >**50**, Neck >**40 cm** and Gender male. A score ≥**3** means high risk; the tool is highly sensitive.
- Ask the **primary motive for getting up** (urge, or already awake). Melatonin and insomnia therapy may help.

> **KEY POINT**
> OSA raises production, lowers capacity and fragments sleep — hence one of the largest nocturia responses to treatment.

7 · Comorbidities

- Comorbidities act mainly through **excess night production**. Nocturia may be the **first sign** of hypertension, metabolic syndrome, coronary, diabetic or early kidney disease.
- EAU: consider systemic causes in all; even **non-bothersome nocturia merits a frequency–volume chart**.
- **Volume overload** (e.g. heart failure): leg fluid is mobilised on lying down and excreted at night.
- **Hypertension**: blood pressure normally dips **10–20%** in sleep. **Non-dipping**, from renin–angiotensin–aldosterone dysregulation, causes **nocturnal natriuresis**.

Disorders of water homeostasis

Arginine vasopressin (AVP) acts on collecting-duct **V2 receptors**; defects give **24-hour polyuria**.

Condition	Defect	Causes
Central diabetes insipidus (DI)	Reduced AVP secretion	Brain tumour, haemorrhage, trauma, autoimmune
Nephrogenic DI	Kidney insensitive to AVP	Drugs, electrolytes, renal failure
Primary polydipsia	AVP suppressed by high intake	Idiopathic; psychiatric

Arginine-stimulated **copeptin** appears superior to water deprivation. Always evaluate global polyuria.

> **KEY POINT**
> Comorbidity acts through the kidney (third-spaced fluid, nocturnal natriuresis, failed antidiuresis). Global polyuria and non-bothersome nocturia both warrant looking beyond urology.

8 · Actions — lifestyle and behaviour

- **Timing matters more than amount**. Water causes diuresis within ≈1 h; caffeine and alcohol within 2–3 h, and they disturb sleep.
- Restrict fluid, caffeine and alcohol from **3–4 h** before sleep; older adults should drink more by day.
- Also: **sodium** restriction, weight loss, exercise, **compression stockings**, bladder training, pelvic floor muscle training (PFMT), treat constipation.

9 · Lower urinary tract dysfunction

- Nocturia accompanies bladder outlet obstruction (BOO), OAB (up to **85%** of trial participants), underactivity, neurogenic bladder and menopause syndrome See Voiding Dysfunction book, Ch 110.
- **Isolated nocturia is not a LUTS**, so it is not expected to respond to bladder or prostate drugs.
- Do urinalysis in all, and **PVR in both sexes**. Check PSA only if it changes management. †Urodynamics are not routine See Voiding Dysfunction book, Ch 107.

Benign prostatic obstruction

- In benign prostatic hyperplasia (BPH) trials, **α-blockers** carry most of the nocturia effect; 5α-reductase inhibitors (5ARIs) alone are weakest. Gains are fractions of a void with a large placebo response: **statistical, not clinical, significance** See Voiding Dysfunction book, Ch 113.
- **Surgery**: TURP beat microwave therapy, α-blockers and watchful waiting; **counsel** that nocturia may respond sub-optimally.

Overactive bladder

- Tolterodine was negative for night voids but reduced urge-related voids; **fesoterodine** (nocturnal urgency endpoint) was positive.
- Patients with a **raised NBCi** respond best to an evening dose. β3-agonists give a marginal benefit (≈0.1–0.2 voids over placebo).
- **Behavioural therapy** comes first and matches or beats drugs. Antimuscarinics are on the **Beers Criteria** for people ≥65 years, the core nocturia population.

Therapy	EAU Male LUTS	EAU Female LUTS
α1-blocker	Offer for nocturia with LUTS (weak)	—
5ARI	Offer with LUTS and prostate > **40 mL** (weak)	—
Phosphodiesterase-5 (PDE5) inhibitor	**Do not offer**	—
Antimuscarinic	Offer with OAB (weak)	Offer after counselling (**strong**)
PFMT; vaginal oestrogen	—	Offer PFMT (**strong**); offer oestrogen (weak)

KEY POINT

LUT drugs act on capacity and outflow, but most nocturia is production-driven — effects are statistically significant, clinically small. Treat the mechanism, not the organ.

10 · Medications causing nocturia

Mechanism	Drugs
Increased urine output	Diuretics; SSRIs (block antidiuretic hormone); **calcium channel blockers** (oedema, raised ANP); tetracycline; lithium (reduces aquaporin-2); **SGLT-2 inhibitors** (glycosuria)
Insomnia / central nervous system effects	Stimulants; α- and β-blockers; salbutamol, theophylline; corticosteroids, thyroxine; SSRIs; phenytoin
Bladder toxicity	Ketamine; tiaprofenic acid; cyclophosphamide

Adjust dose timing first, or substitute another drug.

11 · Management algorithm and refractory nocturia

ALGORITHM

1. Take a history, examine, and obtain a diary and basic tests.
2. Identify the Sleep CALM factors.
3. Optimise each factor, with multidisciplinary help.
4. Add conservative measures.
5. If nocturia stays refractory, then consider an antidiuretic or a timed diuretic in selected patients.
6. Repeat the diary at every step.

- Treat interacting causes together (EAU **"Shared Care Pathway"**). **Target night volume even when capacity is small**.

Desmopressin

P3 Desmopressin (DDAVP) — selective V2 agonist, no pressor effect; hyponatraemia risk factors (age ≥65, Na <135, CrCl <60, high dose), most in first 2–3 weeks; EAU <65 / low dose >65; contraindications

Not in the paper answer:

- Desmopressin is the **only US-approved nocturia drug**, for **≥2 voids per night due to NP** (level 1b); some NP patients have a **blunted nocturnal AVP rise**.
- †Also contraindicated with systemic or inhaled glucocorticoids and estimated glomerular filtration rate (eGFR) <50 mL/min.

	Nasal spray	Orally disintegrating tablet (ODT)
Dose	1.66 mcg if <65 with no risk factors; **0.83 mcg** if ≥65 or at risk	**Sex-specific**: women **25 mcg**; men **50 mcg** (75 mcg added hyponatraemia, not efficacy)
Timing	≈30 min before bed	Sublingual, ≈60 min before bed

- **Women are more sensitive**, because the V2 receptor gene is on the **X chromosome**.
- †Advise minimal fluid from 1 h before until 8 h after the dose.
- Hyponatraemia is defined as sodium < **130 mmol/L**; most cases occur within **3 weeks**.
- **Sodium monitoring** (EAU): baseline, **day 3**, day 7, 1 month, then periodically (US label: baseline, within 7 days and ≈30 days). EAU Female: avoid if baseline sodium is low.
- A **low dose** still works and avoids antidiuresis outlasting sleep. **Age 65 is the pivot.**

Timed diuretics

- Furosemide acts for 6–8 h. Evening dosing **causes** NP, whereas **morning or early-afternoon** dosing treats it (level 1b).
- Small trials (bumetanide 4–6 h, or furosemide 40 mg 6 h before bed) cut voids; monitor sodium.
- **Trap**: afternoon furosemide with bedtime desmopressin worked in a trial, but loop diuretics are a **contraindication** to desmopressin in US labelling.

> **KEY POINT**
> Refractory nocturia: move urine production out of the night. The price is sodium balance — dose, age and monitoring define safe use.

12 · Landmark trials

Trial	Design	Result	Take-home
MTOPS	Doxazosin, finasteride, combination, placebo	Doxazosin and combination beat placebo; finasteride did not	α-blocker drives benefit
VA Cooperative	Terazosin, finasteride, combination, placebo	Only terazosin separated from placebo	5ARI alone does not help
CombAT	Tamsulosin, dutasteride, combination	Combination −0.5 vs −0.3/−0.4 voids	Significant but small
Fesoterodine RCT	Nocturnal urgency primary endpoint	Positive	Helps urgency-driven night voids
Nasal spray phase III	0.83/1.66 mcg vs placebo	Both beat placebo	Basis of approval
ODT sex-specific RCTs	Women 25; men 50/75 mcg	Positive; 75 mcg caused more hyponatraemia	Sex-specific dosing

123 Surgical Procedures for Sphincteric Incontinence in the Male: Artificial Urinary Sphincter and Perineal Slings

1 · Pathophysiology and device mechanisms

- Male **sphincteric (stress) urinary incontinence (SUI)** needs failure of **both** sphincters. Radical prostatectomy (RP), the commonest cause, removes the internal sphincter, leaving the rhabdosphincter.
- Other causes: radiotherapy (RT), pelvic fracture, neuropathy, vesicourethral anastomotic stenosis (VUAS) treatment. See Voiding Dysfunction book, Ch 106.

Device	Mechanism	Site	Adjustable	Needs residual sphincter
Artificial urinary sphincter (AUS, AMS 800)	Circumferential compression, cycled by the patient	Bulbar urethra or bladder neck	No	No
Advance / Advance XP	**Repositions** and lengthens the urethra; no compression	Bulbar, transobturator	No	**Yes**
Virtue (quadratic)	Prepubic compression and elevation	Bulbar, four arms	No	Less dependent
ATOMS (adjustable transobturator male system)	Cushion compression	Bulbar, transobturator	Yes, via scrotal port	Less dependent
ProACT	Two para-urethral balloons	Bladder neck	Yes, via scrotal ports	—

- The **AUS** has a cuff, a scrotal pump and a pressure-regulating balloon (PRB) that sets cuff pressure. The cuff refills passively over minutes.
- A transobturator sling raises leak point pressure **without obstruction** by elevating the bulb and rhabdosphincter.

> **KEY POINT**
> Every device compresses the urethra or repositions a functioning sphincter. Trap: ATOMS and Virtue are "slings" but compress — only the Advance depends absolutely on a coapting residual sphincter.

2 · Indications and device selection

- Indication: **irreversible sphincter incompetence** with bothersome incontinence (UI) after conservative care. Do not delay surgery beyond **6 months** for severe SUI.
- 24-hour pad weight grades severity: mild **<150 g**, moderate **<400 g**. AUS is preferred if >250 g or after RT (MASTER trial See Ch 106).

Stratum (AUA/SUFU)	Recommendation
Severe UI or **any RT**	AUS first-line; sling only with counselling

Stratum (AUA/SUFU)	Recommendation
Moderate, no RT	AUS or male sling; adjustable balloons in specialised centres
Mild, no RT	Continue pelvic floor muscle training (PFMT), sling, possibly AUS

- Counsel: ~1 thin pad/day, AUS lifespan **7–10 years**; failed slings mostly go to AUS.
- **A sling needs** mild–moderate SUI, a contractile bladder (if hypocontractile, choose AUS), a non-irradiated urethra and a coapting sphincter.
- **An AUS needs** manual dexterity and cognition, a healthy urethra and adequate bladder compliance.

Absolute — need an alternative	Relative
Low compliance threatening the kidneys; low capacity; urethra too poor (alternatives: clamp, catheter, bladder neck closure, diversion)	Frequent transurethral work (bladder cancer, refractory VUAS, radiation cystitis), because it risks erosion and infection

- Radiation cystitis needs hyperbaric oxygen and **6 months** free of bleeding before AUS.

> **KEY POINT**
> Each device's contraindication mirrors the other's requirement.

3 · The high-risk ("fragile") urethra

- Strong erosion risk factors are **prior RT, prior erosion or infection, and urethroplasty** (worse than RT, as transection divides the bulbourethral arteries).
- After a prior erosion, 5-year repeat erosion approaches **50%**.
- Countermeasures: optimise medically, use a **transcorporal (TC) cuff**, a PRB of **51–60 cm H2O**, a cuff 0.5 cm larger, and nightly deactivation.

> **KEY POINT**
> Erosion = spongiosal ischaemia under constant pressure. Countermeasures add tissue (TC tunica) or lower pressure — never squeeze harder.

4 · Operative technique

- Sterile urine; give IV antibiotics within **60 min** of incision: aminoglycoside 5 mg/kg plus a first/second-generation cephalosporin or vancomycin.
- Place the cuff on the **proximal bulbar urethra, proximal to corporal fusion**, where spongiosum is thickest.

> **ALGORITHM**
> 1. Midline perineal incision; divide bulbospongiosus.
> 2. Incise Buck fascia and make a 2-cm tunnel around the urethra.
> 3. Measure and place the cuff (usually 4.0–4.5 cm).
> 4. Place the PRB preperitoneally and fill it with **23 mL**.
> 5. Place the pump subdartos in the anterior scrotum.
> 6. Cycle the device; do urethroscopy.
> 7. Leave it **deactivated for 4–6 weeks**.

- Less tight cuffs continent better; a **3.5-cm cuff** raises erosion after RT. †Deactivate the cuff before any later catheterisation or endoscopy.

PRB (cm H2O)	Use
51–60	High-risk bulbar (RT, prior erosion, atrophy); trades continence for safety
61–70	Standard bulbar cuff
71–80	Bladder neck cuff

- **Tandem cuff**: a second cuff ≥1.5–2 cm distal; not first-line, because the distal cuff erodes more.
- **Transcorporal cuff** (Webster 2002): ventral tunica stays on the dorsal urethra and adds bulk. Use it for a urethra <3.5 cm, revisions, prior erosion or urethroplasty; erosion benefit is unproven.
- **Bladder neck AUS** suits an intact prostate (neuropathic bladder) and is **contraindicated after RP**; cuff ≥8 cm, PRB 71–80.

ALGORITHM

1. **Advance XP**:
2. Perineal incision; open bulbospongiosus.
3. Divide the perineal body to release the bulb.
4. Pass the helical trocar outside-in below the adductor longus tendon.
5. Fix the mesh to the spongiosum.
6. Tension under cystoscopy until the mucosa coapts (bulb moves 3–4 cm proximally).

- **ATOMS** leaves bulbospongiosus **intact** and is adjusted in clinic via the scrotal port; **ProACT** balloons are placed at the bladder neck under fluoroscopy.

5 · AUS complications and management

ALGORITHM

1. **Retention**:
2. Confirm the cuff is deactivated.
3. Insert a small catheter for 24–48 h.
4. If the second trial fails, then place a **suprapubic cystostomy**, avoiding the PRB.
5. If retention lasts weeks, then the cuff is undersized: replace it.

- **Infection** occurs in **1–3%** (up to 10% if high-risk), usually staphylococci; **scrotal pain is the key early sign**. If antibiotics fail, explant all; salvage only if not eroded, septic or immunosuppressed.
- **Erosion** occurs in **2–15%** (up to 35% with several risk factors). Remove **all components**. If <50% of the circumference is lost, leave a catheter ~3 weeks; if >50%, do in situ repair or anastomosis.
- Reimplant after **3–6 months** at a **new site**, using a TC cuff with or without a low-pressure PRB.
- **Atrophy** causes a **gradual** recurrent leak. Downsize, relocate (TC cuff distally) or add a tandem cuff; **never simply raise PRB pressure**.
- **Mechanical failure**: within 3 years, replace the failed part; **after 3 years, replace the entire device**.

Recurrent leak after AUS	Clue, then how to confirm
Deactivated	Sudden; reactivate
Fluid leak	Sudden; PRB shrinks on imaging
Atrophy	Gradual; PRB volume normal (>20 mL)
Erosion / detrusor overactivity	Cystoscopy / urodynamics

KEY POINT

Sudden = device, gradual = urethra. Manage every complication to keep a salvageable urethra for the next cuff.

6 · Sling complications and failure

Device	Commonest	Erosion / explant
Transobturator	Transient groin pain; retention or urgency 4–10%	0–0.4%
ATOMS	More complications	Erosion/infection ~10%; explant 10–16%
ProACT	**Perforation** (~5%); migration; revision 22%	~4%

- Cystoscope every worsening sling. After a failed sling the **AUS is the gold standard**, with results as good as primary AUS.

7 · Special circumstances

- A stricture at the cuff may herald erosion.
- With a penile prosthesis, synchronous and staged placement do equally well; if staged, place the AUS first. **Climacturia** responds to a sling or AUS, adding a **mini-jupette** graft if there is erectile dysfunction.

8 · Outcomes

P2 P3 Post-prostatectomy incontinence; SUI: evaluation in females, role of UDS, and surgical therapy in males — AUS social continence 55–77%, satisfaction ~90%, revision 10–29% at 3 years; Advance cure ~60%; ATOMS ~two-thirds dry

Not in the paper answer:

- Cure clusters at 55–65% for all devices (ProACT ~60–67%); Advance XP results are poorer after RT and in severe UI.

★ Putting it together

A · Nocturia: three diary questions, by danger

ALGORITHM

1. **Global polyuria?** If >40 mL/kg/24 h, then investigate for diabetes, diabetes insipidus or polydipsia.
2. **Nocturnal polyuria?** If night fraction >20% (young) or >33% (older), then seek heart failure, sleep apnoea, oedema or evening fluids; consider desmopressin with sodium checks.
3. **Reduced night capacity?** If NBCi >1.3, then treat overactive bladder, obstruction or residual (See Ch 112).

B · Post-prostatectomy incontinence

KEY POINT

Pelvic floor training first; operate on severe leakage by ~6 months. Pad weight: mild <150 g, moderate <400 g. A sling suits mild–moderate leakage, a contractile detrusor and no radiotherapy. Leakage >250 g, radiotherapy or a failed sling needs a sphincter. For erosion risk, add tissue (transcorporal cuff) or lower pressure; never tighten. Remove eroded or infected devices (See Ch 123).